

Comparing Clustering Algorithms for Customer Segmentation

1. Business Problem

Understanding customer behavior is critical for businesses aiming to improve customer retention, increase revenue, and enhance personalized marketing strategies. This paper explores customer segmentation using clustering techniques to categorize customers based on their purchasing behaviors. The key objectives include:

- Comparing different clustering algorithms in segmenting customers based on RFM analysis.
- Evaluating cluster quality to determine the most effective segmentation approach.
- Assessing the impact of clustering on predicting Customer Lifetime Value (CLV) and customer churn.

2. Background

Customer segmentation is a long-established method in marketing and sales, dating back to early direct marketing efforts. Traditionally, businesses used demographic data to segment customers, but modern techniques leverage transaction data to create more precise behavioral segments. With advancements in machine learning, clustering algorithms such as K-Means, Hierarchical Clustering, and Gaussian Mixture Models (GMM) have become instrumental in refining customer segmentation strategies. Several projects have been conducted on customer segmentation; however, none have compared most of the clustering algorithms. As mentioned above, the main objective of this project is to compare different clustering algorithms for customer segmentation based on purchasing behavior using RFM analysis.

3. Data Explanation

3.1. Data Source

The dataset comes from UCI Machine Learning Repository containing transactional data with attributes such as invoice number, product description, quantity, unit price, customer ID, and country. This dataset contains 8 features and 541,909 instances. The dataset contains transactions from a UK-based, registered non-store online retailer, recorded between December 1, 2010, and December 9, 2011. Each row represents a purchase event with attributes including:

- InvoiceNo: Unique identifier for a transaction.
- StockCode: Product identifier.
- Description: Product name.
- Quantity: Number of items purchased.
- InvoiceDate: Date and time of purchase.
- UnitPrice: Price per unit of the product.
- CustomerID: Unique identifier for a customer.
- Country: Country of the customer.

3.2. Data Preparation

Missing values in the CustomerID column were removed to ensure data integrity and accurate analysis. This step was necessary to maintain consistency in customer-related records.

Transactions with negative quantities, which indicate cancellations, were filtered out. Excluding these records helped prevent distortions in the analysis and ensured that only valid purchases were considered. Recency, Frequency, and Monetary (RFM) scores were then calculated for each customer. These scores provided insights into customer behavior by measuring how recently a customer made a purchase, how often they made purchases, and the total monetary value of their transactions.

3.3. Clustering Algorithms Used:

- K-Means Clustering: Partitions customers into K clusters based on RFM features.
- Hierarchical Clustering: Groups customers in a tree-like structure to identify relationships.
- Gaussian Mixture Model (GMM): Probabilistic approach to clustering customers.
- DBSCAN (Density-Based Spatial Clustering of Applications with Noise): Identifies clusters based on density, making it useful for detecting outliers.

3.4. Evaluation Metrics:

The Silhouette Score is used to evaluate the quality of clustering by measuring how well each data point fits within its assigned cluster. It considers both the compactness of the cluster and its separation from other clusters, with higher scores indicating better-defined clusters.

The Calinski-Harabasz Index assesses the dispersion and separation of clusters by comparing the variance between clusters to the variance within clusters. A higher index value suggests that the clusters are well-separated and have distinct groupings.

3.5. CLV and Churn Prediction Models:

For CLV prediction, linear regression-based models used to estimate the total revenue expected from a customer over time. For the Churn prediction, logistic regression and random forests utilized to identify customers at risk of discontinuing purchases.

4. Result

Cluster Characteristics

During the clustering analysis, using K-Means clustering, three customer segments emerged. Cluster 0 consists of frequent but moderate spenders who present a moderate risk of churn. These customers are ideal candidates for retention campaigns aimed at increasing their engagement. Cluster 1 represents high-value customers who purchase less frequently but spend significantly more per transaction. Meanwhile, Cluster 2 includes frequent shoppers whose

spending is moderate, suggesting consistent engagement but room to grow their transaction value.

In the Agglomerative Clustering approach, Cluster 0 identifies occasional high-value customers. These individuals may respond well to targeted incentives that encourage more frequent purchases. Cluster 1 captures frequent, high-value customers with low churn risk, representing a highly loyal and profitable group. On the other hand, Cluster 2 is made up of customers who shop frequently but tend to spend less per visit, also showing moderate churn risk.

For the Gaussian Mixture Model (GMM) Clustering, Cluster 0 includes customers with moderate shopping frequency and moderate transaction value, with a corresponding moderate churn probability. Cluster 1 consists of moderate-frequency shoppers who tend to be high-value spenders and show low churn risk, suggesting a stable and valuable segment. Cluster 2 represents highly frequent shoppers whose spending ranges from moderate to high, with a moderate risk of churn.

Finally, the DBSCAN Clustering method identified two key clusters. Cluster 0 includes customers with moderate frequency and value, facing a moderate churn risk. Cluster 1, however, stands out as a premium group made up of extremely high-value. This segment is crucial for long-term business sustainability and should be prioritized for loyalty and reward programs.

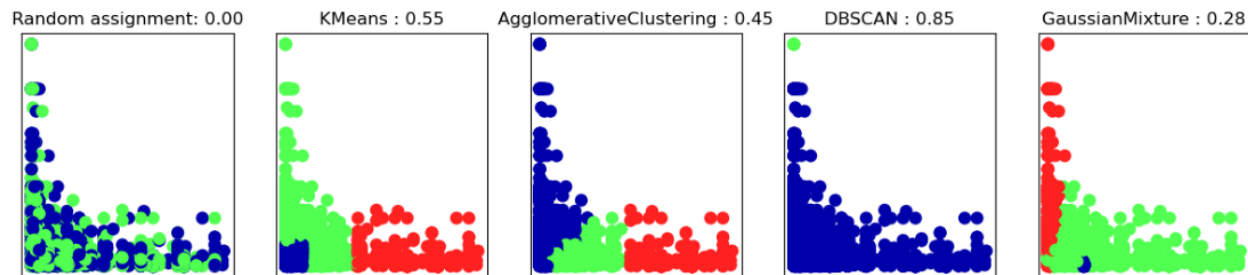
4.1. Clustering Performance:

a. Silhouette Score

The silhouette score evaluates how well-separated clusters are, with values ranging from -1 to 1. Higher scores indicate well-defined and distinct clusters, while lower scores suggest overlapping or poorly separated clusters.

Among the clustering algorithms tested, DBSCAN comes first (0.8534), indicating it has the most distinct and cohesive clusters among all algorithms. Agglomerative and KMeans achieved the second and third highest Silhouette Score 0.4625, 0.04239 respectively, suggesting that they generate relatively well-separated clusters, though not as distinct as DBSCAN. GMM Clustering

had a lower score of 0.2746, suggesting Gaussian Mixture Model (GMM) performed the worst, indicating that its Gaussian-based soft clustering approach did not effectively capture the data structure. The clustering outcome can be seen below on graph-1.



Graph -1: Clustering analysis output using scaled dataset.

b. Calinski-Harabasz Index

The Calinski-Harabasz Index measures cluster compactness and separation, with higher values indicating better-defined clusters. Among the clustering methods tested, Agglomerative achieved the highest score (32508.45), suggesting well-structured and clearly separated clusters. K-Means followed with a score of 27988.34, maintaining a strong cluster structure, this can be seen in table 1.

In contrast, Gaussian Mixture Model (GMM) had a lower score of 14778.89 compared to the K-Mean and Agglomerative, indicating that its clusters were not well-separated. DBSCAN had the lowest score (313.63), which is expected since it does not assume spherical clusters like K-Means. Overall, Agglomerative and K-Means emerged as the best choice, balancing good cluster separation (as indicated by the Silhouette Score) and strong structure (as measured by the Calinski-Harabasz Index).

Table -1: Clustering performance metrics

Clustering Algorithm	Silhouette Score	Calinski-Harabasz Index
KMeans	0.423877	27988.34
Agglomerative Cluster	0.462506	32508.45
GMM	0.274642	14778.89
DBSCAN	0.853389	313.63

While DBSCAN is effective at identifying dense clusters, it may struggle with datasets containing varied-density clusters. GMM performed the worst, suggesting that customer behavior in this dataset is not well represented by a Gaussian distribution.

4.2. Impact on CLV and Churn Prediction:

The Predicted CLV represents the estimated revenue each customer is expected to generate over their lifetime, varying significantly across individuals. Customers with higher monetary values tend to have greater CLV scores, reflecting their long-term revenue potential. For instance, as it can be seen on table 2, Customer 12346 has the highest predicted CLV at approximately \$75,494.84, primarily due to its substantial spending, whereas Customer 12350 has the lowest predicted CLV at \$334.44. This trend highlights the importance of identifying high-value customers and implementing strategies to maximize their engagement and spending.

The Churn Probability metric indicates the likelihood of a customer discontinuing their relationship with the business, with higher probabilities suggesting a greater risk of churn. Table 2 also reveals, Customer 12346 (85%) and Customer 12350 (94%) have high churn probabilities, making them prime targets for retention efforts, while Customer 12348 (98%) is at even greater risk of disengagement. In contrast, Customer 12347 (3%) exhibits strong engagement and is unlikely to churn. Understanding these probabilities allows businesses to proactively address

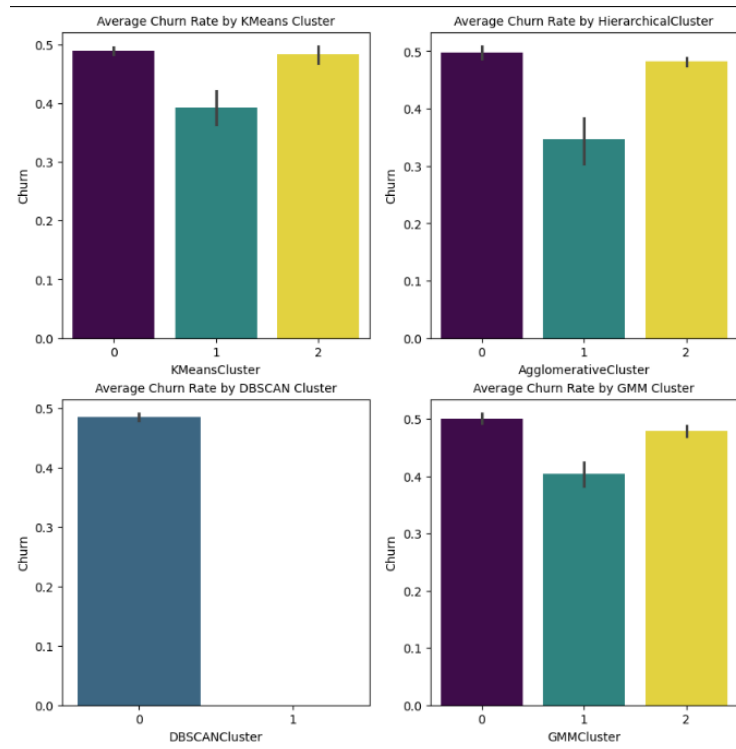
potential risks, ensuring sustained revenue by focusing on customer retention strategies such as personalized offers, loyalty programs, and timely re-engagement initiatives.

Table -2: Customer data with clustering labels, predicted CLV, and churn

Customer ID	Recency	Freq	Monetary	KMeans Cluster	Hierarchical Cluster	DBSCAN Cluster	GMM Cluster	Predicted_ CLV	Churn	Churn Probability
12346	326	1	77183.60	3	4	-1	1	75494.84	1	0.85
12347	2	182	4310.00	1	2	0	0	4311.44	0	0.03
12348	75	31	1797.24	3	4	0	1	1797.56	1	0.98
12349	19	73	1757.55	3	1	0	3	1758.25	0	0.07
12350	310	17	334.40	2	3	0	2	334.44	1	0.94

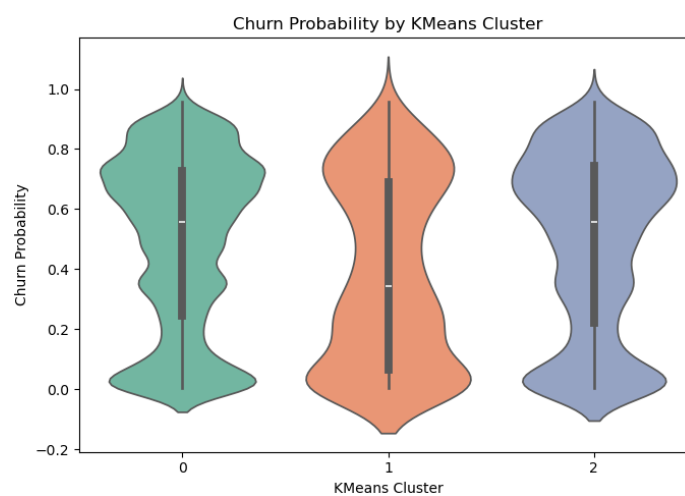
Overall as it can be seen below in graph-2, both K-Means and Agglomerative Hierarchical Clustering consistently identified a cluster with a high average churn rate (labeled 0 in both) and a cluster with a lower average churn rate (labeled 1 in both). This suggests some robustness in these findings. On the other hand, DBSCAN's result of potentially identifying only one main cluster (or classifying others as noise) indicates that the data's density distribution might be such that only one significant dense group exists according to the algorithm's parameters. GMM also identified clusters with varying churn rates, but the distinction between the high-churn clusters (0 and 2) and the lower-churn cluster (1) might need further investigation based on the characteristics of these clusters.

Graph-2: Churn Rate by Cluster



When we see the Churn Probability Distribution by Cluster in graph-3, cluster 1 appears to have the highest overall churn probability compared to the other two. The bulk of the distribution is shifted towards higher churn probabilities, and the median seems to be above 0.6. This suggests that customers in this cluster are generally more likely to churn. This group should be the primary focus of churn prevention efforts.

Graph-3: Churn Probability Distribution by Cluster



5. Conclusion

After evaluating customer segmentation through various clustering algorithms, KMeans, Agglomerative Clustering, DBSCAN, and GMM, distinct customer groups were identified based on RFM (Recency, Frequency, Monetary) metrics, predicted CLV, and churn probability. Each algorithm uncovered unique insights about customer behaviors and value:

- ✓ KMeans and Agglomerative Clustering produced clear and interpretable clusters with strong Silhouette Scores (~0.56 and ~0.46, respectively).
- ✓ DBSCAN revealed a standout group of ultra-high value, loyal customers, although it captured fewer clusters.
- ✓ GMM Clustering identified segments with subtle differences but showed lower Silhouette and Calinski-Harabasz scores, indicating less distinct separation.

The clustering analysis reveals several key customer segments. One prominent group is the high-value, low-churn customers, identified in Agglomerative Cluster 1, DBSCAN Cluster 1, and GMM Cluster 1. These customers purchase frequently and spend significantly, making them major contributors to the predicted Customer Lifetime Value (CLV). They represent a highly valuable segment that should be prioritized for loyalty and retention strategies.

Another important group consists of moderate-value, moderate-churn customers, which appear across most clustering models, notably in K-Means Cluster 0 and GMM Cluster 0. These individuals shop less frequently and have a moderate risk of churning, indicating an opportunity for targeted engagement to increase both retention and transaction value.

Lastly, frequent, low-to-moderate value customers are highlighted in GMM Cluster 2 and Agglomerative Cluster 2. These shoppers buy often but tend to spend less per purchase, suggesting they are volume buyers. This segment may benefit from personalized upselling strategies or product bundling to increase their average transaction value.

6. Assumptions

During the analysis, I made several assumptions to ensure the reliability of insights. The data is assumed to accurately reflect customer purchasing behaviors, serving as a strong foundation for analysis. In addition, no external factors, such as seasonality or economic conditions, were considered to have significantly impacted transactions, ensuring that observed trends are primarily driven by customer actions. Lastly, customer behavior is assumed to remain consistent over time, allowing for meaningful predictions based on historical data.

7. Limitations

A key limitation of this analysis is the absence of demographic information in the dataset, which could improve customer segmentation and provide deeper insights into purchasing behaviors. Additionally, the definition of churn is based solely on recency, which may not fully capture all aspects of customer disengagement, potentially leading to less precise churn predictions.

8. Challenges

In this analysis, several challenges emerged, including determining the optimal number of clusters to accurately segment customers. The other notable challenge faced in the preprocessing phase is handling missing and inconsistent data posed difficulties in maintaining data quality and reliability. Another key challenge was ensuring that the model remains generalizable across different business scenarios, allowing for broader applicability and robust performance in varying contexts. One of the major challenges encountered during this analysis was dealing with memory errors, which often occurred when processing large datasets or running complex machine learning models. I used resample method to reduce the sample size to tackle this issue.

9. Future Uses/Additional Applications

For future applications, the dataset could be expanded to include social media and web browsing behavior data, providing a more comprehensive view of customer interactions. On top of that, incorporating deep learning techniques could also enhance customer segmentation by identifying complex patterns and improving predictive accuracy.

10. Recommendations

Based on the findings mentioned above, I recommend the following actions:

- I. Strengthen Retention of High-Value Segments.
 - ❖ Agglomerative Cluster 1, KMean Cluster 1 and DBSCAN Cluster 1 represent a valuable customer segment. For this group, businesses should consider launching loyalty rewards, early access, or VIP programs to enhance engagement and retention. Offering exclusive discounts and tailored communications can further strengthen customer relationships. Additionally, implementing predictive modeling can help anticipate their needs and personalize offers, leading to improved customer satisfaction and long-term loyalty.
- II. Nurture Moderate-Value, Moderate-Churn Customers
 - ❖ KMeans Cluster 0 and GMM Cluster 0 include customers who may benefit from targeted re-engagement strategies. For this segment, businesses should implement win-back campaigns or personalized product recommendations to reignite interest. Testing email and ad campaigns based on previous purchase timing can help improve engagement. Additionally, promoting bundling or bulk buying incentives can encourage higher transaction values and strengthen customer relationships.
- III. Upsell and Cross-Sell to Frequent, Low-Monetary Shoppers
 - ❖ GMM Cluster 2 and Agglomerative Cluster 2 consist of customers who are ideal candidates for strategies that drive higher profitability. For this group, recommending higher-margin or complementary products can increase overall value. Personalized messaging tailored to their purchase history can enhance engagement and relevance. Additionally, offering volume-based loyalty perks, such as “buy 5, get 1 free”, can incentivize repeat purchases and strengthen brand loyalty.

11. Implementation Plan

The implementation plan involves several key steps: First, ensure that the data is up-to-date and well-structured by following effective data collection and cleaning processes. Next, enhance feature engineering by incorporating both behavioral and demographic attributes to improve the analysis. For clustering analysis, start with K-Means for initial segmentation. Afterward, explore model combination methods to refine and fine-tune the segmentation. Finally, validate the clustering model's effectiveness using historical data to assess its performance and accuracy. Once a fully functioning model is in place, the next step will be deployment and continuous monitoring.

12. Ethical Assessment

In terms of ethical assessment, ensuring data privacy by adhering to relevant regulations and expectations is crucial when handling customer information. In addition to data privacy, addressing bias and fairness is important to prevent discriminatory outcomes in segmentation models. Transparency plays a key role as well, necessitating clear communication about how customer data is utilized to foster trust and uphold ethical standards.

13. References

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